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DEVELOPMENT OF A 3D PRINTED DAMPING SYSTEM FOR NATURAL ROBOTIC HUGS

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Abstract. *Damping systems are widely used in various industries, including aerospace, biomedical devices, and robotics, to control motion and mechanical effects. In social robots, these systems ensure safe and natural physical interactions, allowing robots to provide and receive comfortable human-like hugs. The development of affordable, robust, and interactive robotic systems requires damping mechanisms that are reproducible, adaptable, and efficient. To address these challenges, 3D printing and additive manufacturing have emerged as cost-effective and versatile alternatives for producing robot components and mechanical actuation systems. This study presents the design, simulation, and development of a 3D-printed piston-cylinder damping system for a social robot. The objective was to identify movement mechanisms that improve the robot's ability to bear loads during both soft and high-impact physical interactions. Initially, the damping mechanism was designed and optimized through computer-based simulations, considering the mechanical and viscoelastic properties of the materials. Finite element method (FEM) assessed displacement and stress responses, confirming the piston's resistance to human-applied forces in multiple directions. After computational optimization, a test setup was developed, and the system was then printed by additive manufacturing, using flexible thermoplastic materials, and tested under the application of mechanical force. Subsequently, the complete system was assembled, providing a new structure for the CASTOR robot, with adaptations to the base to improve impact absorption and stability. The results confirm the viability of additive manufacturing in producing efficient and low-cost damping systems for social robots. The adaptability of the printed structure allows design modifications for specific applications. Future research will explore alternative material compositions and sensor integration to monitor performance and real-world applicability.*

Keywords: *Damping system, 3D-printed technology, Robotic hug.*

1. INTRODUCTION

A damping system is a mechanism that dissipates or absorbs energy to prevent oscillations or rigid mechanical responses (Sun and Jing, 2016). Commercially, pneumatic systems are widely used to absorb energy and avoid damaging devices, such as cars, bicycles, bridges, and window systems (Zhou *et al.*, 2016). These mechanical functions are also employed in more complex environments, including industrial automation (Newman and Khoshdarregi, 2022), aerospace (Rittweger *et al.*, 2002), biomedical devices (Romero *et al.*, 2009), and robotics (Tsai *et al.*, 2023). Social robots (SRs) are robotic devices designed to interact socially with humans in communication, education, and healthcare (Cifuentes *et al.*, 2020). These interactions may involve physical human-robot interaction (pHRI), for which several mechanisms have been developed to ensure safety, robustness, and perceived naturalness by users (Casas-Bocanegra *et al.*, 2020). In particular, some applications of social robots include emotionally supportive physical interactions, such as hugging, as well as physical interaction during therapy sessions (Paterson, 2023).

To provide these valuable capabilities, the development of affordable SRs requires actuation systems that enable movement and gestures for hugging, a soft and warm contact surface (Block *et al.*, 2023), and an energy absorption system

to receive the hug naturally and safely, avoiding structural damage to the robot (Gaitán-Padilla *et al.*, 2021). Damping systems are essential to enable such natural and human-like physical interactions. Furthermore, reproducibility and adaptability have become key requirements for damping systems in social robots (Marhefka and Orin, 2002; Papakostas *et al.*, 2018). In response, 3D printing and additive manufacturing have emerged as accessible and versatile alternatives to produce robot components (Casas-Bocanegra *et al.*, 2020) and mechanical actuation systems (Krause and Bhounsule, 2018). These manufacturing technologies allow developers to create integrated, lightweight, and customizable solutions that improve the natural behavior of low-cost robots (Paterson, 2023).

Various materials used in 3D printing offer different mechanical characteristics, including tensile strength, hardness, flexural strength, and elongation (Iftekar *et al.*, 2023). Common polymers include acrylonitrile butadiene styrene (ABS), a durable and high-temperature-resistant low-cost material; polylactic acid (PLA), a biodegradable and easy-to-print material; polyethylene terephthalate glycol (PETG), which offers durability and flexibility; and thermoplastic polyurethane (TPU), suitable for elastic and flexible applications (Prajapati *et al.*, 2024). Despite advances in physical human-robot interaction (pHRI), the development of systems capable of providing safe, natural, and affective contact, such as robotic hugs, remains limited by the high cost, mechanical complexity, and lack of customizability of existing solutions. Many commercial damping systems are not easily adaptable to the irregular forces and contact dynamics typical of human gestures like hugs, especially in SR applications.

To address these limitations, this study proposes a low-cost, reproducible, and adaptable damping system, developed through additive manufacturing and tailored for integration into a SR. The system was designed to absorb and dissipate forces from multidirectional physical interactions, providing a mechanical support that enhances safety and naturalness during robotic hugs. By testing different materials and geometries, and conducting simulations and mechanical tests, the developed system was integrated into an SR structure. This work contributes to the growing field of human-centered robot design by introducing a 3D-printed piston-cylinder damping mechanism that supports pHRI, advancing the physical robustness for emotional comfort of a SR.

2. METHODOLOGY

This section describes the methodology combining the mechanical design, simulation, and additive manufacturing techniques to develop a safe and natural physical interaction mechanism for robotic hugging. The process began with the adaptation of the robot's mechanical structure to integrate the new damping system, followed by the design and development of a flexible piston-spring mechanism capable of absorbing forces. The Finite Element Method (FEM) was used to simulate the system's behavior under varying force conditions to select the optimal geometry and material. The simulations were extended to the robot's structure with applied forces in the external pieces to validate the developed system. Finally, the selected spring design was 3D printed and analyzed with observational functioning through a developed set-up of the mechanical structure of the robot.

2.1 Mechanical structure of the 3D printed robot

The mechanical structure of the robot was designed to enable safe and natural physical interactions, particularly in hugging scenarios. The development of the cushioning system's mechanical structure involved computational modeling, numerical simulations, and additive manufacturing prototyping (3D printing). The primary materials used were TPU, PLA, PETG, and ABS, selected for their flexibility and structural integrity. The main objective was to design a robust structure capable of absorbing impact forces and sustaining loads during physical interactions while maintaining mechanical durability. The computational modeling and simulation of the mechanical structure were performed using SolidWorks 2024, a computer-aided design software (CAD).

Firstly, the support structure for the main damping cylinder was developed, followed by the adaptation of the back part of the CASTOR robot's base for the stability foundation, as depicted in Figure 1 (a). Subsequently, topological optimization was performed to enhance the balance between rigidity and weight.

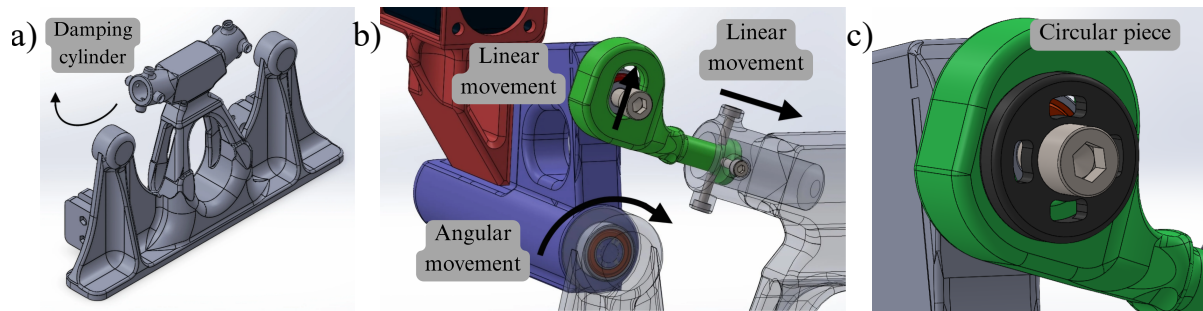


Figure 1. Mechanical adaptations to the base structure. (a) The back of the robot's base with the main damping cylinder. (b) Passive movement mechanism with the rotational axis in green to convert it into linear motion inside the cylinder. (c) Designed circular component to restrict the movement of the rotational-to-linear part.

A mechanism was required to convert rotational to linear motion while imposing a movement amplitude constraint. To achieve this, the piece presented in green in Fig 1 (b) was designed, the groove allows a degree of freedom for relative vertical movement between the piece and the bearing (SKF 626-2Z). This configuration results in rotary motion at the joint and linear motion in the damping cylinder. To complement the motion mechanism, modifications were made to the lateral part of the robot's structure, shown in Fig 1 (b), incorporating a hole to secure the bearing (SKF 626-2Z) with a screw (M6x35 A2-70) in its upper section.

Additionally, since the screw head was insufficient to restrict the translation of the green part along the orthogonal axis of the piece's plane, a circular component was designed to constrain this movement, as presented in Figure 1 (c).

Considering the weight of the components of the robot's arms, passive joints naturally tend to rotate outward, necessitating a restriction on the system's range of motion. To address this, mechanical stops with screws (M3x10) were implemented within the main damping cylinder, making a slight diameter variation in the green piece. The Fig 2 (b) shows the new parts adapted to the system: the back part in gray, the side part in purple, the vertical part of the arms in red, and the rotational-to-linear part in green.

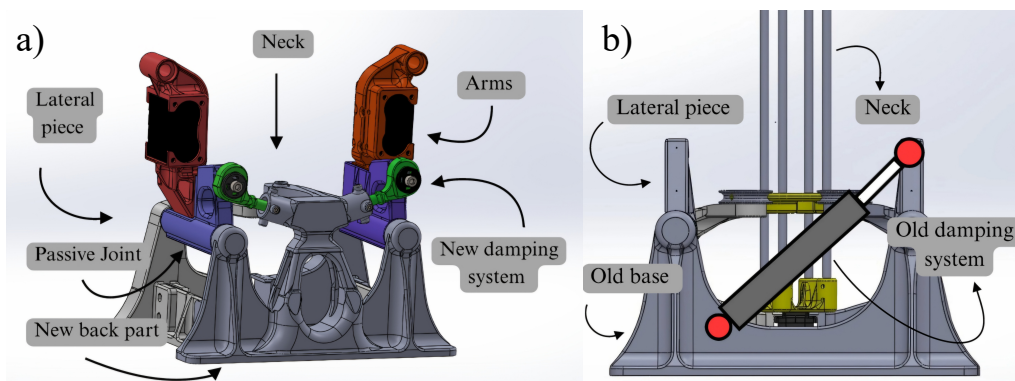


Figure 2. The base of the CASTOR robot. (a) Old version with a pneumatic commercial piston. (b) New version with the 3D printed damping system.

2.2 Damping system development

For the design of the damped spring, parameters such as geometry, material (TPU, PLA, PETG, and ABS), wall thickness, and internal filling patterns were considered to optimize energy dissipation. Three different geometries were developed: a helical spring with a rectangular profile, a helical spring with a circular profile, and a spring hose, presented in Figure 2 (a).

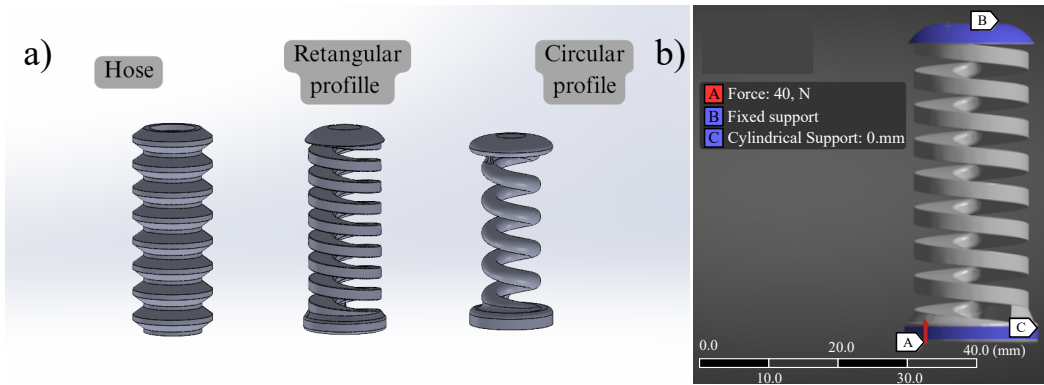


Figure 3. The damped spring. (a) The three geometries studied. (b) Static simulation of a helical spring with a rectangular profile.

2.3 Damping system simulation

Subsequently, computational simulations were performed using the FEM, with the Mechanical 2025 R1 software (ANSYS, Inc). Initially, a structural behavior evaluation was conducted under different loading and impact conditions to select the best geometry and material. The main aspects analyzed included stress distribution, strain distribution, and fatigue analysis under the applied loads.

Static structural simulations were performed to evaluate different spring geometries and materials. Each simulation was configured as shown in Figure 2 (b), with fixed support at the upper end (B), and a cylindrical support applied to the outer edges of the component (C), constraining radial movement while allowing axial and tangential displacements. A compressive force (A) was applied to assess deformation and stress distribution under realistic operating conditions.

2.4 Force analysis in the robot's structure

To validate the mechanical strength of the system's main structural components, static structural simulations were carried out with loads applied. To simplify the simulations and reduce the computational load, it was divided into two parts: the back part and the motion mechanism (made up of the arm part, side part, rotational-to-linear piece, bearing, and screw). The printed parts were considered to be made of PLA.

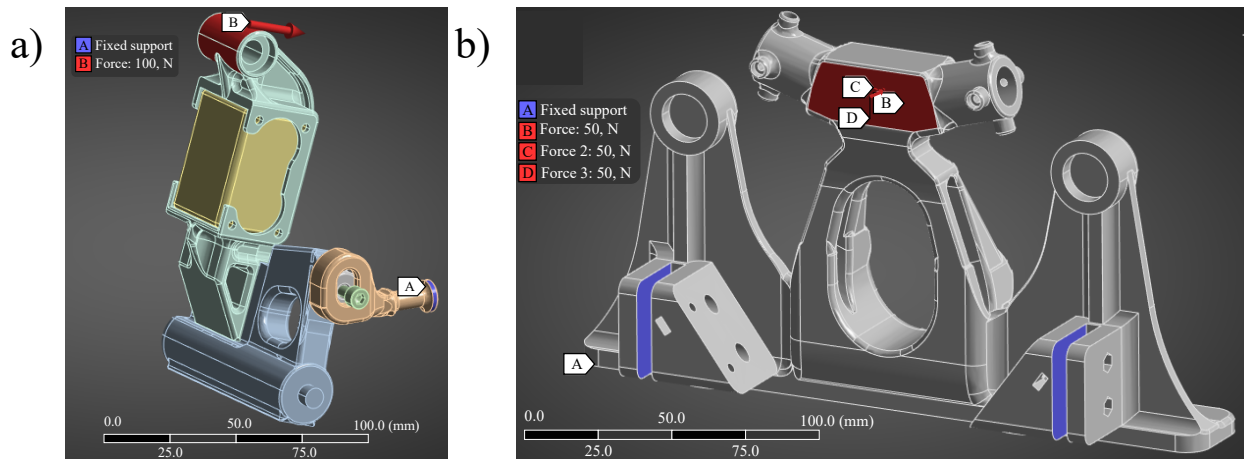


Figure 4. Structure simulation. (a) Static simulation of the motion mechanisms. (b) Static simulation of the back part.

In the simulation of the back part, the lower face in contact with the ground and the faces in contact with the other stabilization parts were fixed, as marked as (A) in Figure 3 (a). Three forces of 500 N were applied, two to the main inner cylinders, and one orthogonal to the red face.

For the main structural simulation of the components forming the motion mechanism, presented in Figure 3 (b), a simplified bearing part was made and assembled for the moment of motion, when the bending moment of the conversion piece reached it's maximum. A simplified motor part (DYNAMIXEL XM540-W270-R) was also made, where the contact areas between it and the vertical arm part were considered to be bonded. The bearing and screw were considered as structural steel. The end of the conversion piece, which would be in contact with the spring of the damping system, was clamped, and the ends of the side piece on the Z axis were considered to be a fixed joint, allowing free rotation on the Z

axis. Finally, a force of 265 N was applied to the upper part of the vertical arm piece, presented in red.

2.5 Flexible spring print

To manufacture the springs using additive manufacturing (3D printing), a Creality K1 Max printer equipped with a 0.4 mm nozzle was employed. To maximize the compressive strength of the components, the springs were printed in a vertical orientation, aligning their main axis with the Z axis of the printer. The slicing process was performed using Creality Print 6.0 software. A 100 % infill density was selected to replicate the assumptions made in the FEM simulations, which considered the parts to be fully dense. Due to the relatively small size and complex geometry of the springs, the internal volume was insufficient for the infill pattern to fully develop and display its expected mechanical behavior, such as uniform stress distribution or enhanced elasticity.

2.6 Observational validation

Finally, for the validation test, an experimental mechanical setup was developed, Figure 10 (a), based on the new parts designed for the robot. The setup for the experiment was printed in PLA, Figure 10 (b). It was then attached with screws (M6X35) to an optical breadboard, so that two manual forces could be applied separately, (1) a fast, immediate force and (2) a 5-second maintained force, both exerted to the maximum compression of the damping system. The spring to be used will be the one with the best results in the static simulations. The observational analysis will be subjective to the expected behavior of the system, considering the simulations, the spring would need to return to its original position and resist the exerted force in both cases.

3. RESULTS

The results of this study are presented in three subsections, (1) the adaptation of the mechanical structure, (2) the analysis of the damping system, and (3) the observational outcomes from the setup. First, the integration of the damping system into the robot's base required structural modifications to ensure stability and compatibility with the flexible spring. Next, mechanical simulations demonstrated the effectiveness of the damping mechanism in dissipating force under various loading conditions, validating the structural integrity of the design and its capacity to absorb energy, and obtaining the best geometry and material for the proposed piston. Finally, observational tests, including repeated force applications and maintained interactions were analyzed for a future validation with users during hugging physical interaction in the robot's structure.

3.1 Mechanical structure

The static structural simulation of the back part, the main part for stabilization, showed a stress field well below the PLA's yield strength, concentrated in the stabilization rods that connect the main cylinder of the shock absorber system to the old CASTOR base part, and a maximum stress of 27.937 MPa, also located in the support rods, as shown in Figure 5 (a). The total deformation results showed very low indications, with a maximum deformation of 0.032 mm, located in the upper part of the part, as this is the region the farthest away from the fixed region, causing the largest moment, as shown in Figure 5 (b).

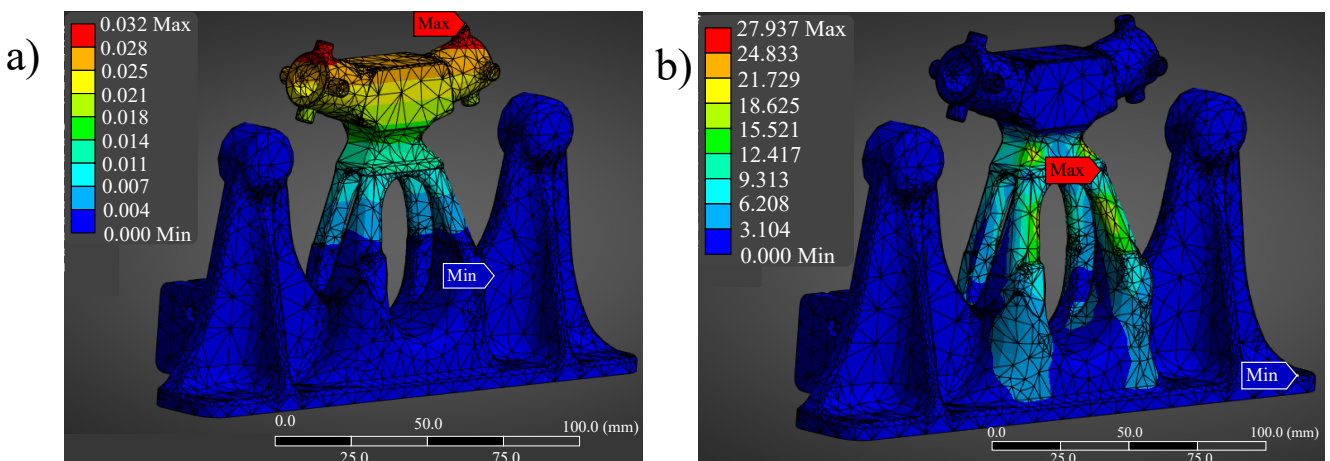


Figure 5. Back part static structural simulation results (a) Equivalent stress (von-Mises) - MPa (b) Total deformation - mm.

The simulation of the movement mechanism showed a maximum stress of 268.60 MPa, but the region with the highest stress was concentrated at the point of contact between the screw and the bearing. The other stress regions had lower levels and were lower than the PLA yield stress, as shown in Figure 6.

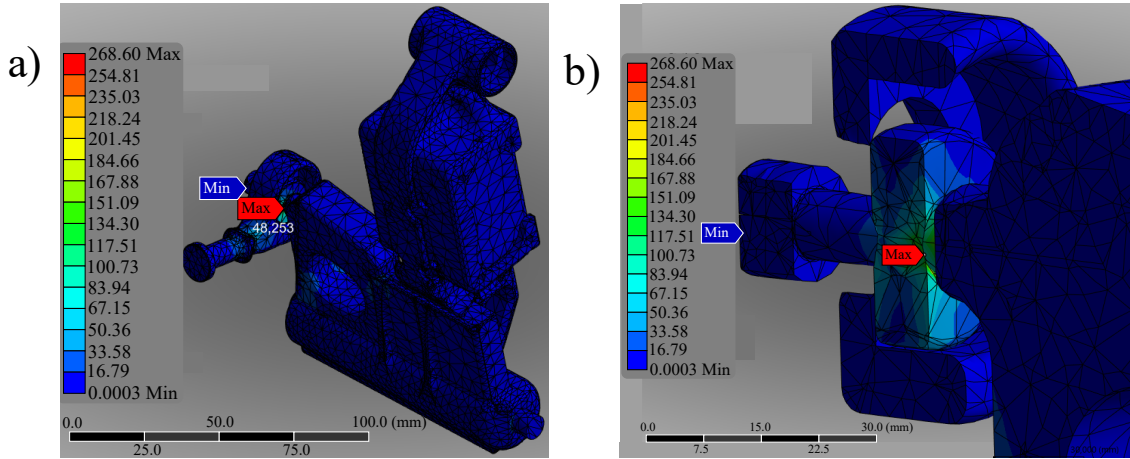


Figure 6. Motion mechanism static structural simulation results. Equivalent stress (von-Mises) (a) Complete view (b) Section plane view.

3.2 Damping system analysis

Three geometries and different combinations of the four materials were analyzed, taking into account the results of Equivalent Stress, Total Deformations, and the fatigue fracture Safety Factor. The Safety Factor is a dimensionless metric that indicates how safe a component is against failure, such as fatigue or rupture. It is defined as the ratio between the material's capacity (e.g., fatigue strength or yield strength) and the applied stress (maximum stress or cyclic loading). A value greater than 1 indicates that the component can withstand the load with a margin of safety, while values below 1 suggest a risk of failure. The results for the CR-TPU hose spring, the PLA rectangular profile spring, and the ABS circular profile spring will be presented below, as they were the best materials for each respective geometry.

The hose springs with the best results, resisting a compressive force of 60 N, in CR-TPU. It showed a low stress field, concentrated in a few small internal regions of the spring and with a maximum stress of 7.37 MPa, as shown in the Figure 7 (a); a maximum displacement of 15.351 mm indicating a good range for the movement of the system, shown in Figure 7 (b); and a fatigue fracture safety factor of 2.71, indicating that the part can withstand almost three times the applied stress before failure, Figure 7 (c).

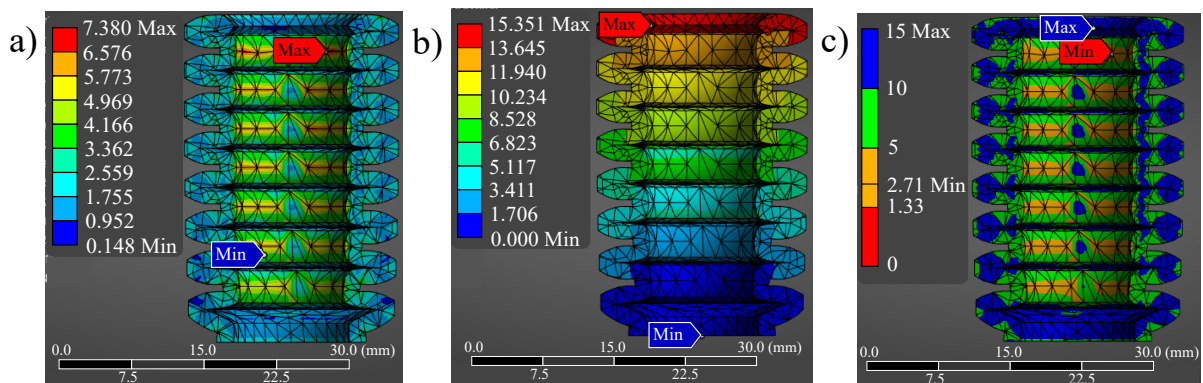


Figure 7. Spring hose static structural simulation results (CR-TPU). (a) Equivalent stress (von-Mises) - MPa, (b) Total deformation - mm, and (c) Fatigue fracture safety factor.

The rectangular spring was able to withstand 20 N in PLA without exceeding its yield strength. Showing a high stress field on the inside of the spring links, seen in Figure 8 (a), with 46.582 MPa of maximum stress. Only 3.542 mm of displacement, indicating a low amplitude for the movement of the system, Figure 8 (b), and a fatigue fracture safety factor of 0.859, showing that the part would fatigue fracture after a few cycles, Figure 8 (c).

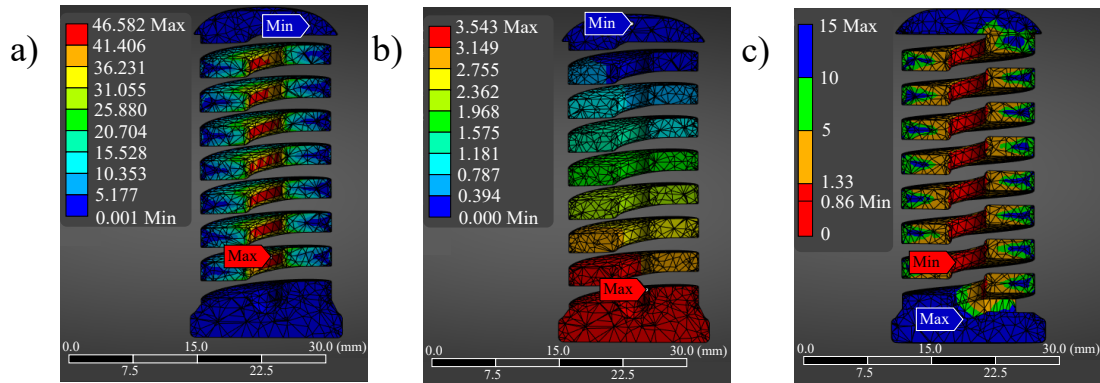


Figure 8. Spring rectangular profile static structural simulation results (PLA). (a) Equivalent stress (von-Mises) - MPa (b) Total deformation - mm (c) Fatigue fracture safety factor.

The circular-profile spring, which exhibited the poorest performance, withstood 12 N in ABS without exceeding its yield strength. A stress concentration field was observed in the inner region of the spring links, with a maximum stress of 34.707 MPa, Figure 9 (a); and a displacement of 3.156 mm, indicating a limited range of motion for the system, Figure 9 (b). The fatigue fracture safety factor was 1.44, indicating a safety margin over repeated loading cycles, Figure 9 (c).

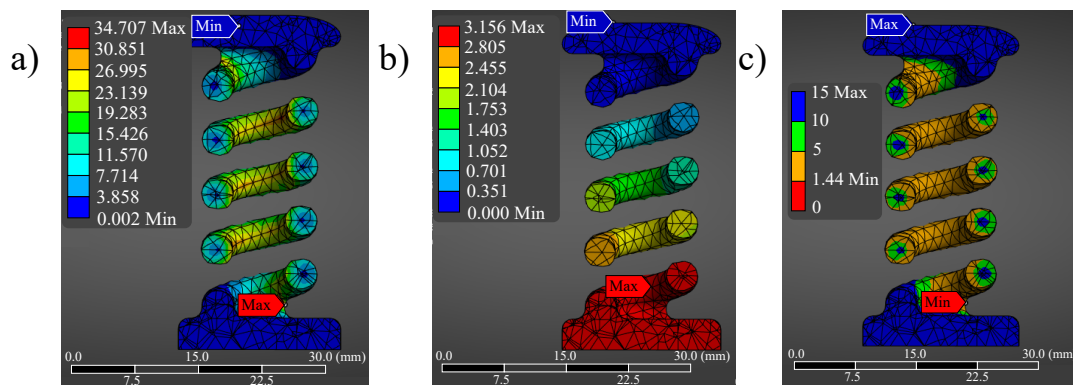


Figure 9. Spring circular profile static structural simulation results (ABS). (a) Equivalent stress (von-Mises) - MPa (b) Total deformation - mm (c) Fatigue fracture safety factor.

3.3 Observational

For the observational evaluation of the printed mechanical setup, which was subjected to two forces, one applied immediately and the other 5 second maintained, both showed good damping and quick response, returning to the initial state when the force was removed, as shown in Figure 10 (c) and (d). As for the movement mechanism, it worked well, with fluid movement and low friction. There was only a certain backlash in the lower bearing, which, because of the base piece made for the test setup, had a small displacement orthogonal to the plane of movement.

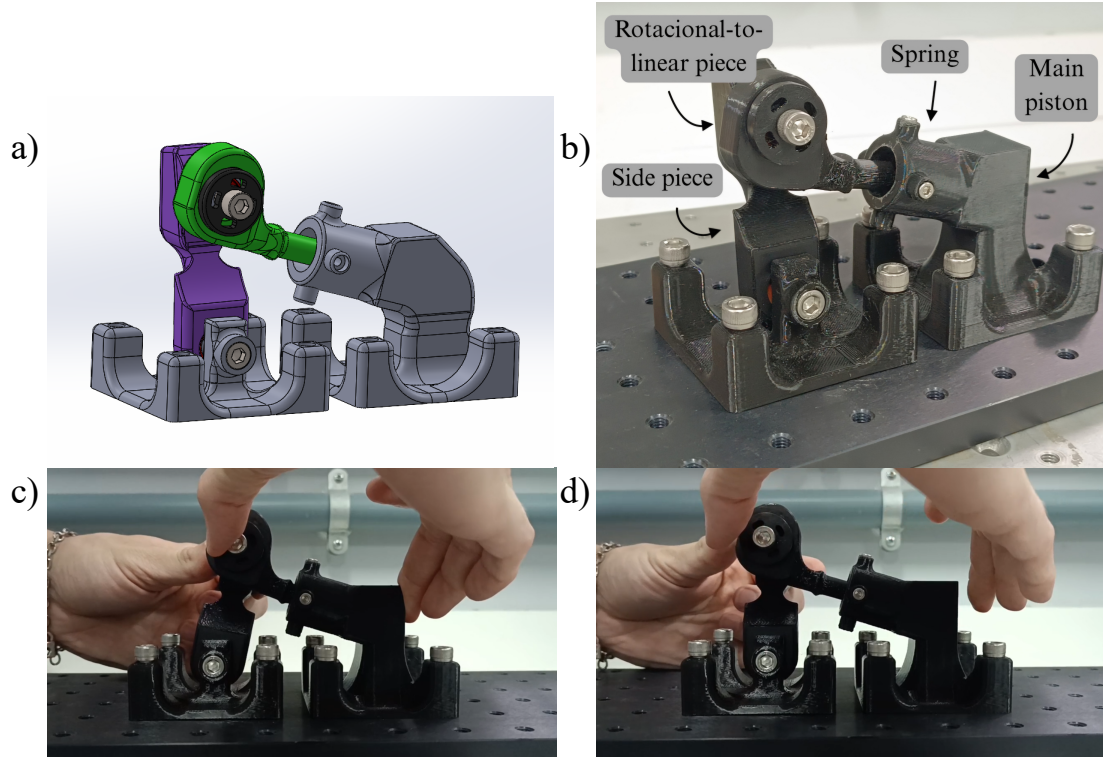


Figure 10. Experimental setup. (a) CAD, (b) Printed setup, (c) Compressive force applied, and (d) Compressive force released.

4. DISCUSSION

The structural parts analyzed showed good performance under forces of relatively high magnitude, characteristic of the context of physical interaction between man and robot in systems with plastic components (Zhou *et al.*, 2020). Considering the yield strengths of the materials used, the A2-70 grade bolt, an austenitic stainless steel, has a yield strength of 450 MPa in tension and 270 MPa in shear (60 % of the tensile strength). However, the PLA polymer material has a yield strength of 50 MPa. The applied stresses remained below the respective yield limits of the materials, guaranteeing the integrity of the system and preventing failures due to permanent deformation.

Concerning damping, based on the results obtained in the static analyses, the best choice of geometry and material was the CR-TPU spring hose, as it managed to reach the expectation of 60 N; a common value in commercial pneumatic dampers, presenting low stress levels and satisfactory deformation for the movement of the damping system and absorption of impact energies.

In contrast, the other geometries and materials showed high stresses, close to the yield limit, even with much lower forces. These stress levels also resulted in a low factor of safety against fatigue fracture for the options not chosen, indicating that, after a certain number of cycles, the parts would tend to fracture due to fatigue, resulting in a shorter service life for the part, as seen in (Holzweber *et al.*, 2018). On the other hand, the CR-TPU spring hose had a high factor of safety against fatigue fracture, indicating a good safety margin and a longer lifetime.

In addition, the spring's compression range of approximately 15 mm proved to be reasonably good for moving the arms in a physical human-robot interaction, although it is smaller than that of commercial pneumatic devices, which can move up to 45 mm.

One of the primary limitations of this study was the high computational cost associated with the FEM simulations. The modeling of the mechanical behavior of the damping system, under realistic physical interaction forces, required detailed and high-resolution meshing. These demands increase simulation time and hardware requirements, which were performed on a workstation equipped with an 8-core processor and a licensed version of Mechanical 2025 R1 software (ANSYS, Inc). Despite this setup, the complexity of the simulations involving the entire robotic structure needed simplifications in geometry or segmentation of the simulation into sub-components. Consequently, this may have limited the fidelity of some stress distribution results and fatigue predictions.

Another limitation lies in the physical reproduction of the damping components using additive manufacturing. Although the 3D-printed parts were designed based on the assumptions of full infill and ideal material homogeneity, practical constraints such as insufficient infill development led to discrepancies between the simulated and the real mechanical behavior. In particular, the 100 % infill setting could not fully reproduce the internal patterns, such as gyroid or honeycomb, reducing the uniformity and elasticity of the predicted stress distribution. Additionally, certain fine structural

features observed in the simulation were difficult to replicate with precision using the 0.4 mm nozzle of the Crealty K1 Max printer, which could impact real-world performance and consistency.

5. CONCLUSION AND FUTURE WORK

This study presented the design, simulation, and development of a 3D-printed damping system for natural robotic hugs. The system was designed to enhance the safety and comfort of human-robot interactions, providing a solution for a low-cost social robot. Through a combination of computational simulations and experimental tests, the damping system demonstrated effective energy absorption and mechanical stability under repeated force applications.

The results indicated that the CR-TPU spring hose was the most suitable configuration, offering optimal energy dissipation and mechanical durability. In contrast, other tested geometries and materials showed higher stress levels, indicating lower reliability under prolonged use. The integration of the damping system into the CASTOR robot's structure was successful, maintaining its functional integrity and natural movement capabilities. Finally, the experimental setup shows the energy absorption capacity of the proposed damping system when real forces are applied.

However, the study also revealed limitations, such as the computational cost of the FEM simulations and the challenges of achieving uniform infill in 3D-printed parts with complex geometries. Future work should explore alternative materials, optimize the geometry of the damping components, and study dynamic movements in the setup using a servo motor and optical fiber sensors to measure and monitor the damping system behavior.

6. ACKNOWLEDGEMENTS

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